

Discussion section 7

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1 Chp 3, Q46

Given a Markov chain with transition matrix P and stationary distribution π , the time reversal is a Markov chain with transition matrix $\tilde{P}$ defined by

$$\tilde{P}_{ij} = \frac{\pi_j P_{ji}}{\pi_i}, \quad \text{for all } i, j.$$

- (a) Show that a Markov chain with transition matrix P is reversible if and only if $P = \tilde{P}$.
- (b) Show that the time reversal Markov chain has the same stationary distribution as the original chain.

Proof. (a) If P is reversible, then $P_{ij} = \frac{\pi_j P_{ji}}{\pi_i}$, for all i, j , i.e., $P = \tilde{P}$.
Conversely, if $P = \tilde{P}$, then $P_{ij} = \tilde{P}_{ij} = \frac{\pi_j P_{ji}}{\pi_i}$, i.e., P is reversible.

- (b) Assume the state space has size n . For each fixed j , summation over index i , in the equations $\pi_i \tilde{P}_{ij} = \pi_j P_{ji}$, we have:

$$\sum_{i=1}^n \pi_i \tilde{P}_{ij} = \sum_{i=1}^n \pi_j P_{ji} = \pi_j \left(\sum_{i=1}^n P_{ji} \right) = \pi_j \cdot 1 = \pi_j.$$

This is exactly $\pi \tilde{P} = \pi$. (Here we are using P is a stochastic matrix, i.e., each row of P add up to one.)

□

2 Chp 3, Q48

Consider a Markov chain with transition matrix

$$P = \begin{matrix} & \begin{matrix} a & b & c \end{matrix} \\ \begin{matrix} a \\ b \\ c \end{matrix} & \begin{pmatrix} 1-\alpha & \alpha & 0 \\ 0 & 1-\beta & \beta \\ \gamma & 0 & 1-\gamma \end{pmatrix} \end{matrix},$$

where $0 < \alpha, \beta, \gamma < 1$. Find the transition matrix of the time reversal chain (see Exercise 3.46).

Proof. Drawing a graph. It's clear that the Markov is finite and irreducible, thus will have a unique stationary distribution π .

We can just solve the stationary distribution π by hand. $\pi P = \pi$ gives the following system:

$$\begin{cases} \pi_a \cdot (1 - \alpha) + \pi_c \cdot \gamma = \pi_a, \\ \pi_a \cdot \alpha + \pi_b \cdot (1 - \beta) = \pi_b, \\ \pi_b \cdot \beta + \pi_c \cdot (1 - \gamma) = \pi_c, \end{cases} \quad (2.1)$$

which gives $\alpha\pi_a = \beta\pi_b = \gamma\pi_c$. Combining with the normalization condition $\pi_a + \pi_b + \pi_c = 1$ we have,

$$\pi = \left(\frac{\frac{1}{\alpha}}{\frac{1}{\alpha} + \frac{1}{\beta} + \frac{1}{\gamma}}, \frac{\frac{1}{\beta}}{\frac{1}{\alpha} + \frac{1}{\beta} + \frac{1}{\gamma}}, \frac{\frac{1}{\gamma}}{\frac{1}{\alpha} + \frac{1}{\beta} + \frac{1}{\gamma}} \right).$$

The transition matrix of the time reversal chain is then

$$\begin{pmatrix} 1 - \alpha & 0 & \alpha \\ \beta & 1 - \beta & 0 \\ 0 & \gamma & 1 - \gamma \end{pmatrix}.$$

□

3 Chp 4, Q7

Give the probability generating function for an offspring distribution in which an individual either dies, with probability $1 - p$, or gives birth to three children, with probability p . Also find the mean and variance of the number of children in the fourth generation.

Proof. The probability generating function is

$$G(s) = \sum_{k=0}^{\infty} s^k \mathbb{P}(X = k) = s^0 \cdot (1 - p) + s^3 \cdot p = (1 - p) + s^3.$$

The mean μ of the offspring distribution is

$$\mu = \sum_{k=0}^{\infty} k a_k = 0 \cdot (1 - p) + 3 \cdot p = 3p.$$

The variance σ^2 of the offspring distribution is

$$\sigma^2 = (1-p) \cdot (0-\mu)^2 + p \cdot (3-\mu)^2 = (1-p)\mu^2 + p(3-\mu)^2 = 9p(1-p).$$

(An even more convenient way is simply using the formulae in section 4.3, i.e., $\mathbb{E}[X] = G'(1)$, and $\text{Var}(X) = G''(1) + G'(1) - (G'(1))^2$).

By the computation of section 4.2 in the textbook, the mean of the 4th generation size $\mathbb{E}[Z_4]$ is

$$\mathbb{E}[Z_4] = \mu^4 = (3p)^4.$$

The variance of the 4th generation size is

$$\text{Var}(Z_n) = \sigma^2 \mu^{4-1} \cdot \frac{\mu^4 - 1}{\mu - 1} = 9p(1-p)(3p)^3 \cdot \frac{(3p)^4 - 1}{3p - 1}.$$

□

4 Chp 4, Q8

If X is a discrete random variable with generating function G . Show that

$$\mathbb{P}(X \text{ is even}) = \frac{1 + G(-1)}{2}$$

Proof. It simply comes from adding up $G(1)$ and $G(-1)$. By definition,

$$G(s) = \sum_{k=0}^{\infty} s^k \mathbb{P}(X = k).$$

Thus

$$1 = G(1) = \sum_{k=0}^{\infty} (1)^k \cdot \mathbb{P}(X = k) = \mathbb{P}(X \text{ is even}) + \mathbb{P}(X \text{ is odd}),$$

and

$$G(-1) = \sum_{k=0}^{\infty} (-1)^k \cdot \mathbb{P}(X = k) = \mathbb{P}(X \text{ is even}) - \mathbb{P}(X \text{ is odd}).$$

Adding up the above two equations and dividing by 2, we obtain the desired result. □

5 Chp 4, Q14

A branching process has offspring distribution with $a_0 = p$, $a_1 = 1 - p - q$, and $a_2 = q$. For what values of p and q is the process supercritical? In the supercritical case, find the extinction probability.

Proof. $\mu = 0 \cdot p + 1 \cdot (1 - p - q) + 2 \cdot q = 1 - p + q$. By definition, the process is supercritical if $\mu > 1$, so $1 - p + q > 1$ has to hold, i.e., $p < q$.

We compute the probability generating function

$$G(s) = s^0 \cdot p + s^1 \cdot (1 - p - q) + s^2 \cdot q = p + s(1 - p - q) + s^2 q.$$

By theorem 4.2 in the textbook, the extinction probability is the smallest positive root of the equation $s = G(s)$. We just solve it:

$$s = p + s(1 - p - q) + s^2 q.$$

This is a quadratic equation in s with two roots $s_1 = p/q$, $s_2 = 1$. In the supercritical case we have $p < q$, so $p/q < 1$. The extinction probability is then p/q .

□